Christian Garry

Stochastic Analysis · Mean-Field FBSDEs · Numerical & Neural Methods | C++/Python
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Publications

Well-posedness of fully coupled McKean–Vlasov FBSDEs with jumps under full-tuple law dependence 2026
- *C. Feng, C. Garry.* Submitted to *Stochastic Processes and their Applications*; arXiv:2608.23203. Existence, uniqueness and stability under jump-extended monotonicity, with coefficients depending on the joint law of (X, Y, Z, U).

A posteriori error certification for McKean–Vlasov FBSDEs with jumps 2026
- *C. Feng, C. Garry.* A two-sided, computable, reference-free residual estimator that brackets the true discrete L^2 error from solver output alone; recovers Reisinger–Stockinger–Zhang (2024) in the no-jump limit.

Martingale Completion for Time-Discrete Stochastic Solvers 2026
- *C. Feng, C. Garry.* Manuscript. Omitting the orthogonal Kunita–Watanabe martingale leaves a fixed-grid error floor that training cannot remove; predictable low-rank completion heads remove it (96.9% median reduction).

Education

MSc Scientific Computing & Data Analysis (AI for Engineering) Durham, United Kingdom
Durham University Sep 2025 – Sep 2026
- Taught-module average **84.75%** (95% GPU programming, 94% ML & statistics); dissertation mark pending.
- **Cohort-highest mark on both MATH52115 summatives**: Regression 92 (mean 85), Classification 87 (mean 69).
- **Dissertation:** *McKean–Vlasov Forward–Backward Stochastic Differential Equations with Jumps* — well-posedness, a posteriori error estimation and martingale completion (the three papers above); supervised by Prof. C. Feng.

MEng Electronic Engineering Durham, United Kingdom
Durham University Sep 2020 – Jun 2024
- Class II.1 (2:1). **Dissertation** (Silicon Carbide JFET CPU) **84%**; standout modules: Solid Mechanics 98, Electronic & Electrical Systems 95.

Mathematics Modules (NetMath) Online, USA
University of Illinois Urbana-Champaign Apr 2026 – Dec 2026
- Credit-bearing, exam-assessed proof-based mathematics (studied alongside the MSc): **MATH 314** Introduction to Higher Mathematics — **A**−; **MATH 447** Real Variables *(in progress)*.

Experience

Software Developer/Engineer Hebburn, United Kingdom
Siemens PLC Sep 2026 – Present

Graduate Communications Engineer Hebburn, United Kingdom
Siemens PLC Sep 2024 – Aug 2026
- Built a **deep-learning high-impedance-fault detector** deployed on a protection device.
- Implemented C/C++ comms components (TCP/IP, serial) between simulated systems and PC tools; built a Python RAG pipeline (local LLMs, vector search) over internal docs.

Industrial Tutor Durham, United Kingdom
Durham University, Department of Engineering Sep 2025 – Present
- Weekly undergraduate design tutorials; rubric-based assessment and structured technical feedback.

Selected Projects

`fbsdej-certify` / `mvfbsdej` **— certified FBSDE solver software** 2026
- *Sole developer.* A posteriori error certificate for any MV-FBSDEJ solver output, plus a tested PyTorch solver library reproducing all paper results. github.com/cgarryZA/fbsdej-certify

Jane Street Advent of FPGA — Hardcaml (OCaml) Dec 2025 – Jan 2026
- Synthesizable FPGA accelerators in Hardcaml (OCaml) for Jane Street's Advent of FPGA. github.com/cgarryZA/advent-of-camel-2025

Silicon Carbide JFET CPU Durham, United Kingdom
Durham University Oct 2023 – Apr 2024
- Independently designed and simulated a 4-bit CPU in SiC JFET logic (LTspice) with a bespoke **C++ compiler/assembler toolchain** and Python waveform automation; validated instruction-level behaviour via emulator parity checks. Mark 84%; **invited speaker**, NEIMME Semiconductors Conference (2024).

Key Skills

Mathematics: stochastic analysis & BSDEs; probability & statistics; numerical analysis; optimisation; measure-theoretic probability.
Computing: Python (NumPy/PyTorch/CUDA), C/C++; HPC & GPU; reproducible scientific software; Linux, Git, CI.
Awards: Weldon le Huray Fencing Scholarship; President's Award (Gold) — SA Duke of Edinburgh.
Interests: Jane Street puzzles.